

Baljinder Singh

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Professional Summary

Passionate Data Science and Machine Learning Engineer skilled in Python, data preprocessing, feature engineering, and model development. Experienced with Scikit-learn, TensorFlow, Keras, Generative AI (LangChain, Transformers), FastAPI, MLOps, and Computer Vision, with strong skills in NLP. Solid foundation in DSA, OOPS, and core CS concepts, focused on building scalable and accurate data-driven solutions.

Skills

- Programming Languages** — Python, C++
- Generative AI** — Transformers, LLM Applications, Prompt Engineering | LangChain, Hugging Face Transformers | RAG Pipelines, Embeddings, Semantic Search | Vector Databases (FAISS, Chroma,Qdrant,Pinecone)
- Agentic AI** — AI Agent Design, Task Planning, Tool Integration | LangGraph Workflow Development | n8n Automation | Multi-step Autonomous Workflows & Memory-based Agents,Crew AI
- Machine Learning And Deep Learning** — Supervised & Unsupervised Learning | Regression & Classification | Neural Networks (ANN, CNN, RNN, LSTM) | CNN-based Image Classification & Transfer Learning | NLP (Embeddings, NLTK, SpaCy) | FastAPI, MLOps
- Cloud Computing** — AWS EC2, AWS S3 Storage, FastAPI Deployment, EC2 Instance Configuration, Basic Auto Scaling
- Databases** — Microsoft SQL Server,MySQL
- Soft Skills** — | Problem Solving | Analytical Thinking | Team Collaboration

Professional Experience

- Generative AI Intern, Coder Roots** 01/2026 – Present
- Contributed to the development of Generative AI and Agentic AI projects using LangChain, FastAPI, and automation workflows.
 - Worked on AI-powered applications involving RAG pipelines, vector databases, API integration, and backend optimization.
 - Collaborated with the development team on project implementation, debugging, testing, and performance enhancement of scalable AI solutions.
- Data Science & Machine Learning Trainee, O7 Services** 06/2024 – 12/2024
- Completed 6 months of training in Data Science and Machine Learning with hands-on experience in Python, data analysis, model building, and AI/ML project development.

Projects

- Medilex-AI — Agentic Medical RAG Chatbot**
- Built an AI-powered medical assistant using React.js, FastAPI, LLMs, Agentic AI, and RAG with Qdrant Vector DB.
 - Implemented symptom analysis, medicine comparison, and context-aware chatbot responses using embeddings and vector search.
- Nexus AI -RAG Assistant**
- Built Nexus AI RAG Assistant that allows users to upload PDFs and extract contextual knowledge for intelligent question-answering.
 - Developed a FastAPI backend integrated with a RAG model to process documents and generate accurate context-based responses.
- Computer Vision Projects**
- Sign Language Recognition System** — Built using MediaPipe; created custom dataset, implemented real-time hand tracking, based on american sign language .
 - Driver Drowsiness Detection System** — Real-time eye and yawn detection using MediaPipe face landmarks with live monitoring and visualization dashboard.
- AI Doctor- Automated Healthcare Web App**
- Developed an AI-powered multi-disease prediction system capable of detecting conditions related to the brain, lungs, and other major organs using trained ML models.

Education

- B.Tech - Computer Science and Engineering,** DAV Institute of Engineering and Technology, Jalandhar, Punjab 2022 – 2026
CGPA-8.78
- 12th-Non Medical Science,** Indo Swiss International Convent School, Nakodar 2021 – 2022
93%
- 10th-,** Indo Swiss International Convent School ,Nakodar 2019 – 2020
86.8%

Achievements

- LeetCode Problem Solving:** Solved 155+ Data Structures & Algorithms problems on LeetCode and earned the 50 Days Continuous Coding Badge
- NEC IIT Bombay:** Participated in the National Entrepreneurship Challenge, contributing to team-based problem solving.. |
- Hackathons:** Participated in multiple hackathons focused on AI, Machine Learning, and software development.